Robust Cointegration Testing for Statistical Arbitrage

Estimator uncertainty, structural change, and sequential inference

Why does $\hat{\beta}_t$ move?

Why does the evidence move?

Thesis progress — September 2026

The presentation starts from two observed instabilities

Problem 1: the estimated parameter fluctuates

$$\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3, \ldots$$

Main explanations

- sampling / estimation uncertainty;
- genuine parameter change;
- model misspecification in the background.

Problem 2: statistical evidence fluctuates

$$T_1, T_2, T_3, \ldots \qquad p_1, p_2, p_3, \ldots$$

Main inferential issues

- repeated looks at the same null;
- strongly dependent repeated statistics;
- several hypotheses monitored together.

First understand **why the quantity moves**; then ask how to make a statistically valid decision from that movement.

Problem 1: why does the estimated coefficient move?

$$\boxed{\hat{\beta}_t = \beta_t + u_t}$$

Hence

$$\boxed{\Delta\hat{\beta}_t = \underbrace{\Delta\beta_t}_{\text{genuine structural change}} + \underbrace{\Delta u_t}_{\text{sampling / estimation fluctuation}}}$$

Sampling variability

β_t may be constant, but a finite sample gives a noisy estimate.

Structural instability

The underlying relationship itself changes over time.

Conditional on an adequate model specification. Misspecification is a third, separate modeling issue.

Source I: sampling / estimation uncertainty

Suppose the underlying relation is locally fixed:

$$Y_t = \alpha + \beta X_t + \varepsilon_t, \qquad \beta_t \equiv \beta.$$

Different nearby samples can still give

$$\hat{\beta}(W) \neq \hat{\beta}(W').$$

Question

- How precisely is β identified in the current sample?
- How sensitive is $\hat{\beta}$ to a plausible perturbation of that sample?

Natural uncertainty measures

$SE(\hat{\beta})$, confidence intervals, bootstrap uncertainty, leverage / information content.

Here we are **not yet testing for a break**. We first ask whether the estimate is fragile even under a locally stable relationship.

Current empirical direction: use SE as a fragility signal

In simple regression,

$$SE(\hat{\beta}) = \sqrt{\frac{\hat{\sigma}^2}{\sum_t (x_t - \bar{x})^2}}.$$

large $SE(\hat{\beta})$ → weakly pinned-down slope → large nearby-window sensitivity

For fixed window length w,

$$S^{\text{move}}_{t,w} = \frac{1}{H}\sum_{h=1}^{H}\left|\hat{\beta}_{t+h,w} - \hat{\beta}_{t,w}\right|, \qquad S^{\text{win}}_{t,w} = \frac{1}{|K|}\sum_{k\in K}\left|\hat{\beta}_{t,w+k} - \hat{\beta}_{t,w}\right|.$$

These are measures of observed estimator instability, not direct measures of structural change.

Existing result: SE predicts estimator fragility

Exploratory rolling experiment: RB=F on HO=F, log-prices, 2015–2026, window lengths 40–240 days.

0.564
$\mathrm{corr}(SE, S^{\mathrm{move}})$

0.679
$\mathrm{corr}(SE, S^{\mathrm{win}})$

Predictor	AUC move	AUC window
$SE(\hat{\beta})$	**0.807**	**0.852**
p-value	0.635	0.638
$-\lvert t\rvert$	0.689	0.710

Interpretation: SE is useful as a qualification / fragility signal. But future $\hat{\beta}$ movement can still contain both sampling fluctuation and true structural change.

Source II: genuine parameter instability / structural change

Now allow the underlying relation itself to vary:

$$Y_t = \alpha_t + \beta_t X_t + \varepsilon_t.$$

After accepting a reference relation β_0, the monitoring question is

$$H_0 : \beta_t = \beta_0 \text{ throughout monitoring} \qquad \text{vs} \qquad H_1 : \text{a parameter change occurs.}$$

Names in statistics

parameter instability; structural break; change point; parameter drift; regime change.

Typical approaches

CUSUM; recursive-residual monitoring; structural-break tests; sequential change detection.

A break detector asks whether the observed movement is **too large or persistent to be explained by sampling variability under H_0.**

Related literature: Horváth et al. (2004, 2007); Kejriwal & Perron (2010a, 2010b).

Cointegration-specific structural-change literature

There is already a direct literature on monitoring cointegrating relationships.

Sequential monitoring

- Wagner & Wied (2015, 2017): monitor stationarity and cointegration after a calibration period;
- Trapani & Whitehouse (2020): monitor slope change and/or breakdown to non-cointegration.

Time-varying parameter models

Eroğlu, Miller & Yiğit (2022) study time-varying cointegration with a Kalman-filter/state-space formulation and highlight potential spurious-regression issues.

This literature addresses the **true-change side** of the parameter problem; the SE work addresses the **sampling-uncertainty side**.

Problem 2: why does the test statistic / p-value move?

Even if the hypothesis itself is unchanged, the statistic changes when the sample changes:

$$T_t = T(\mathcal{W}_t), \qquad p_t = p(\mathcal{W}_t).$$

For a rolling window,

$$\mathcal{W}_t = (X_{t-w+1}, \ldots, X_t) \quad \longrightarrow \quad \mathcal{W}_{t+1} = (X_{t-w+2}, \ldots, X_{t+1}).$$

sample changes $\rightarrow$ T_t changes $\rightarrow$ p_t changes

The fluctuation itself is normal. The inferential problem starts when we **repeatedly inspect** these changing statistics and act when one crosses a threshold.

Repeated looks: the same null is tested again and again

Suppose we repeatedly inspect the same question:

$$H_0, \qquad H_0, \qquad H_0, \qquad \dots$$

with $p_1, p_2, p_3, \dots$.

Problem names

repeated significance testing; repeated looks; sequential monitoring; optional stopping.

Error phenomenon

Type-I error / false-alarm probability can exceed the nominal single-look level.

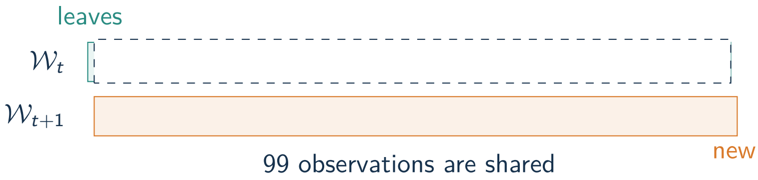
The question is no longer "is $p_t < 0.05$ now?" but "what is the probability that I **ever** falsely trigger while monitoring?"

Related literature: Horváth et al. (2004, 2007) for sequential monitoring; Trapani & Whitehouse (2020) for cointegrating regressions.

Overlapping windows: a second low p-value mostly reuses old evidence

Let $w = 100$.

$$\mathcal{W}_t = (X_{t-99}, \ldots, X_t), \qquad \mathcal{W}_{t+1} = (X_{t-98}, \ldots, X_t, X_{t+1}).$$



If $p_t = 0.02$ and $p_{t+1} = 0.02$, the second result is **not an independent replication**: only one observation entered and one left.

Central question: how much genuinely new evidence is contained in the update from $\mathcal{W}_t$ to $\mathcal{W}_{t+1}$?

Response to repeated, dependent monitoring: use sequential evidence

There are two broad strategies.

Keep rolling tests

Calibrate the *joint* null distribution of the repeated statistics, e.g. through sequential boundaries / max-statistic theory / suitable resampling.

Build one sequential evidence process

Condition on the past and let only newly arriving information update the evidence:

$$\mathbb{E}_0[e_{t+1} \mid \mathcal{F}_t] \leq 1, \qquad E_{t+1} = E_t e_{t+1}.$$

This second viewpoint formalizes the intuition: **do not count the 99 shared observations as fresh evidence again**.

Related literature: Classical route: Horváth et al. (2004, 2007), Wagner & Wied (2017), Trapani & Whitehouse (2020). E-process motivation: Hartog & Lei (2025); Kim et al. (2026).

A different issue: we monitor more than one hypothesis

At a fixed monitoring time t we may ask two distinct questions:

Coefficient hypothesis

$$H_t^{\beta} : \ \beta_t = \beta_0.$$

Has the accepted coefficient relation changed?

Residual hypothesis

$$H_t^{\varepsilon} : \ \varepsilon_t \text{ remains stationary.}$$

Does the residual relation still hold?

Testing both creates a **family of hypotheses**. As an independent illustration,

$$P(V \geq 1) = 1 - (1 - \alpha)^m,$$

so two independent tests at $\alpha = 5\%$ give $P(V \geq 1) = 9.75\%$.

Problem name: multiple testing / multiplicity. The overall false-positive criterion is often the **family-wise error rate (FWER)** $P(V \geq 1)$. Dependence changes the numerical formula, but the family still needs simultaneous error control.

More subtle: the two hypotheses are structurally dependent

The residual hypothesis is defined using the coefficient:

$$\varepsilon_t(\beta) = Y_t - \alpha - \beta X_t.$$

Therefore

$$H^{\varepsilon}(\beta_0): \ \varepsilon_t(\beta_0) \text{ is stationary}$$

is a different hypothesis from

$$H^{\varepsilon}(\beta_1): \ \varepsilon_t(\beta_1) \text{ is stationary.}$$

test coefficient H^{β} → update $\beta_0 \to \beta_1$ → residual hypothesis is redefined

This is deeper than correlated p-values: **one monitoring decision changes the statistical object that the other hypothesis refers to**.

After a β update: what happens to accumulated stationarity evidence?

Suppose

$$\hat{\beta}_0 \longrightarrow \hat{\beta}_1.$$

Then

$$\hat{\varepsilon}_t^{(0)} = Y_t - \hat{\alpha}_0 - \hat{\beta}_0 X_t \quad \longrightarrow \quad \hat{\varepsilon}_t^{(1)} = Y_t - \hat{\alpha}_1 - \hat{\beta}_1 X_t.$$

Evidence collected for $\hat{\varepsilon}^{(0)}$ is **not automatically evidence** for $\hat{\varepsilon}^{(1)}$.

evidence for $\varepsilon^{(0)}$ → update β → new residual $\varepsilon^{(1)}$ → continue? discount? restart?

Specific research problem: evidence management after adaptive model re-estimation.

Related literature: Related adaptive-switching perspective: Kim et al. (2026). Time-varying cointegration caution: Eroğlu, Miller & Yiğit (2022).

After a β update: what can happen to old evidence?

Logical hierarchy – not a competing solution. Residual evidence for $\varepsilon(\beta_0)$ is interpreted only while β_0 is the maintained coefficient relation. Replacing β_0 changes the residual hypothesis itself.

After $\beta_0 \to \beta_1$, three genuine alternatives remain:

1. Full restart

Stop evidence for $\varepsilon(\beta_0)$, re-qualify, and start a fresh process for $\varepsilon(\beta_1)$.

Cleanest baseline; loses old evidence.

2. Evidence transfer / discounting

Retain some old information under a rule that accounts for the changed residual definition.

More efficient; validity is non-trivial.

3. Joint adaptive monitoring

Use one sequential procedure in which β_t adapts and new evidence is conditional on the past.

Most integrated; technically harder.

Research question: can pre-update information be reused without treating evidence for $\varepsilon(\beta_0)$ as evidence for $\varepsilon(\beta_1)$?

Related literature: Related directions: adaptive switching of e-processes (Kim et al., 2026) and e-value based error control (Hartog & Lei, 2025).

Final thesis structure

Observed $\hat{\beta}$ movement

Sampling uncertainty
SE / estimator fragility

Structural change
break / change-point monitoring

Observed evidence movement

Repeated looks
overlap and sequential validity

Multiple hypotheses
FWER / simultaneous control

Adaptation links the two sides: changing β redefines the residual hypothesis
restart | hierarchical / gatekeeping | joint adaptive sequential evidence

Working message: separate estimator uncertainty from genuine change; control repeated and multiple inference; then decide how evidence should behave when the model itself is updated.

References I: structural change and cointegration monitoring

- Horváth, L., Hušková, M., Kokoszka, P., & Steinebach, J. (2004). *Monitoring changes in linear models*. Journal of Statistical Planning and Inference, 126, 225–251.
- Horváth, L., Kokoszka, P., & Steinebach, J. (2007). *On sequential detection of parameter changes in linear regression*. Statistics & Probability Letters, 77(9), 885–895.
- Kejriwal, M., & Perron, P. (2010a). *Testing for Multiple Structural Changes in Cointegrated Regression Models*. Journal of Business & Economic Statistics, 28(4), 503–522.
- Kejriwal, M., & Perron, P. (2010b). *A sequential procedure to determine the number of breaks in trend with an integrated or stationary noise component*. Journal of Time Series Analysis, 31(5), 305–328.
- Wagner, M., & Wied, D. (2015). *Monitoring Stationarity and Cointegration*. SSRN Research Paper 2624657.
- Wagner, M., & Wied, D. (2017). *Consistent Monitoring of Cointegrating Relationships: The US Housing Market and the Subprime Crisis*. Journal of Time Series Analysis, 38(6), 960–980.
- Trapani, L., & Whitehouse, E. (2020). *Sequential monitoring for cointegrating regressions*. arXiv:2003.12182.
- Eroğlu, B. A., Miller, J. I., & Yiğit, T. (2022). *Time-varying cointegration and the Kalman filter*. Econometric Reviews, 41(1), 1–21.

References II: repeated testing, multiplicity, and adaptive evidence

- Tian, J., & Ramdas, A. (2021). *Online control of the familywise error rate*. Statistical Methods in Medical Research, 30(4), 976–993.
- Hartog, W., & Lei, L. (2025). *Family-wise Error Rate Control with E-values*. arXiv:2501.09015.
- Kim, H., Wu, S., Venkatraman, A., Lin, G., & Kim, S. (2026). *Deciding When to Switch: E-Processes for Adaptive Minimax Training for Generative Adversarial Nets*. arXiv:2608.10096.
- Perumal, K., & Flint, E. (2018). *Systematic testing of systematic trading strategies*. Journal of Investment Strategies, 7(3).

Positioning: the classical econometric papers provide the structural-change / cointegration-monitoring backbone; the online-FWER and e-value papers provide tools for repeated and multiple inference; the adaptive-switching paper is a recent related example rather than a cointegration-specific solution.